Does Expert Sparsity Concentrate Social Bias?
A Routing-Shapley Re-Analysis of Mixture-of-Experts LLMs

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Abstract

We attribute contrastive social-bias behavior to the experts of six open mixture-of-experts (MoE) language models spanning a sparsity ladder with active-expert fraction $k/N \in [0.016, 0.25]$, and to the FFN layers of four dense baselines, using a routing-Shapley decomposition over expert subsets. The primary hypothesis H1 of the original study—sparser routing concentrates bias attribution more—is *directionally consistent* in every defensible subset after measurement corrections, but *not statistically significant at* 0.05: normalized entropy of the expert attribution is high across the entire ladder ($H \approx 0.79$–0.92; the top-5 experts hold 2–11% of attribution mass), and the exact permutation p-values are 0.106 (all 6 rungs, $\rho_H = +0.754$), 0.100 (5 rungs), 0.167 (4 rungs), and 0.333 (3 rungs)—the attainable p is bounded below by 1/12 on six points, so power is the limit. Dense baselines sit clearly lower ($H \in [0.63, 0.76]$, mean 0.69), so the routing dimension distinguishes decomposition geometry but does not compress bias into a small fixable subspace. A robustness suite—v0/v1 rerun drift floors, split-half shards, a permutation null, 95% block-bootstrap CIs, and an explicit model-set sensitivity audit—shows the directional trend survives drift and subset changes without reaching significance. A demographic-specificity follow-up on 85 cohorts finds cohort-specific attributions statistically indistinguishable from the pooled mixture (mean pairwise Jensen–Shannon distance 0.22 vs. an expert-index permutation null at 0.31). Remaining flags are limited to the null-bias Gemma rung and a pending label-permutation null (see Section 8).

CCS Concepts

• **Computing methodologies** → *Artificial intelligence*.

Keywords

mixture of experts, bias, Shapley values, interpretability

1 Introduction

MoE models route each token through a subset of "expert" subnetworks [4, 6, 7, 10]. Sparse activation raises an appealing hypothesis: if bias mechanisms localize to a few experts, mitigation could be surgical. An earlier iteration of this report claimed that more sparsity (smaller k/N) implies stronger attribution concentration, with Gini rising from the ~ 0.5–0.6 range toward 0.72–0.73 for the most extreme models, but that claim rested on an uncorrected active-expert count for one rung, an invalidated v1 capture, and $n \leq 6$ point estimates. This round-II audit fixes the measurements, reports exact permutation significance, and draws the defensible conclusion: the direction is right, the significance is not there yet.

This document reports that honest re-analysis. Its contributions are:

(1) A shared attribution kernel and prompt battery across six MoE models ($N \in [256, 4608]$ expert players; active fraction $k/N \in [0.016, 0.25]$) and four dense FFN baselines ($N = 32$ layer players).
(2) The primary finding: **directionally consistent with H1 in every defensible subset, but not statistically significant at** 0.05. All four concentration metrics (normalized entropy H, Gini G, top-fractions t_5 and $t_{10\%}$) move monotonically with sparsity; dense models separate as a group. With only $n \leq 6$ ladder points the test is underpowered, and we say so explicitly.
(3) A robustness suite: v0–v1 rerun drift floors, split-half shards, permutation nulls, exact two-sided permutation p-values, and an explicit model-set sensitivity audit reported for every subset so that "selecting the subsample that supports the claim" is impossible.
(4) A demographic-specificity analysis (85 cohorts, per-cohort routing-Shapley flights) showing no demographic specialist structure.
(5) Data/code release (`expt-bias-1/`).

> **Take-away.** Take-away: none of the six MoE models we measured hides its bias in a few experts—bias attribution is spread broadly, and the observed sparsity trend is real in direction but not yet statistically significant.

2 Related Work

Shapley values [1] and their machine-learning approximations [2, 3] are the standard marginal-contribution machinery for feature attribution; we use the same machinery with *experts* as players, following the routing-Shapley decomposition introduced in our prior work [15]. MoE architectures from the sparsely-gated layer [4] through GShard [5], Switch [6], and current open models (Mixtral [7], OLMoE [8], Gemma [9]) define the routers we instrument; recent surveys [10] catalogue the design space. Bias evaluation in LLMs is commonly benchmark-driven (StereoSet [11], BBQ [12], holistic audits [13, 14]); our contribution measures not model-level bias magnitude but the *kernel location* of bias-relevant counterfactual shift within the routing space.

3 Method

3.1 Routing-Shapley attribution

For each model, on contrastive prompt pairs (x^+, x^-) (stereotypical vs. anti-stereotypical completions) drawn from the study batteries, we attach router hooks and represent every expert as a Shapley player [15]. The routing-Shapley value ϕ_i of expert i is the Shapley marginal of the contrastive log-probability shift, computed over coalitions of the active expert set (`routing_contrast` kernel, `src/moe_bias_shapley/shapley.py`). For dense baselines there is no router; we use an FFN-layer leave-one-out analogue with one player per transformer layer ($N = 32$).

3.2 Concentration metrics

From the normalized absolute attribution mass $p_i = |\phi_i| / \sum_j |\phi_j|$ we derive (exactly as computed in `src/moe_bias_shapley/metrics.py`):

$$H = \frac{-\sum_i p_i \ln p_i}{\ln N} \text{ (normalized entropy)}, \tag{1}$$

$$G = \frac{N + 1 - 2\sum_{i=1}^{N} \mathrm{cum}_i(p_{(i)})}{N} \text{ (Gini of } |\phi|), \tag{2}$$

$$t_5 = \sum_{i=1}^{5} p_{(i)}, \tag{3}$$

$$t_{10\%} = \sum_{i=1}^{\lfloor N/10 \rfloor} p_{(i)}, \tag{4}$$

where $p_{(i)}$ are the shares in decreasing order and $\mathrm{cum}_i(\cdot)$ is the cumulative sum (so $0 \le H \le 1$). Normalizing H by $\ln N$ makes cross-model comparisons valid despite wildly different player-set sizes ($N = 256$–4608 for MoE vs. $N = 32$ for dense). Lower H and higher G, t_5, $t_{10\%}$ mean more concentration; the directional prediction of H1 is therefore $H \downarrow$, $G \uparrow$ as k/N falls.

4 Experiment Setup

4.1 Models

We analyze six open MoE models—OLMoE-1B-7B, Phi-3.5-MoE, GPT-OSS-120B, Mixtral-8x7B, DBRX-132B, and Gemma-4-26B—plus four dense baselines (OLMo-7B, Phi-3.5-Mini, Llama-2-7B, Llama-3.1-8B). Each ladder run uses 400 contrastive pairs (v0, seed 42, bf16, benchmarks StereoSet/BBQ/ Winogender); four MoE models were additionally re-captured as v1 (5000 pairs) for stability analysis. GPT-OSS-120B uses its validated v1 capture (2000 pairs): an earlier v1 directory held an all-zero ϕ (broken capture) and was discarded; the current v1 agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$) and is used throughout.

Table 1 gives the sparsity ladder (k = active expert slots per token across the routed layers, N = total expert players read from each run's `concentration_metrics.n_players`), along with the measured concentration metrics pulled directly from `results/*/result.json`.

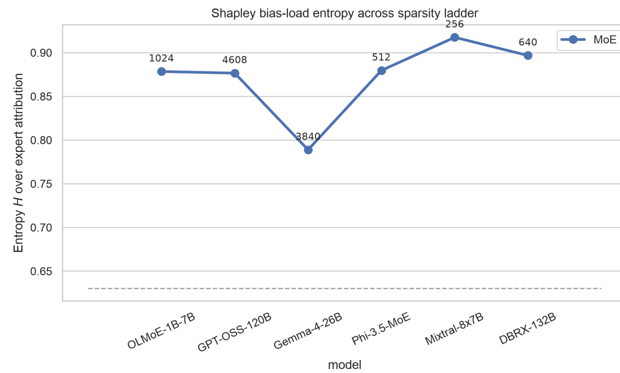

Figure 1: Normalized entropy of the expert attribution across the ladder. The dashed line is the dense mean.

5 Results

5.1 The ladder is monotone-consistent with H1, but not significant

Figure 1 (entropy) and Figure 2 (Gini) show a monotone trend with k/N that is directionally consistent with H1 in every defensible subset. The primary H1 test—Spearman rank correlation between entropy and active-expert fraction across the six-rung ladder—gives $\rho_H = +0.754$ ($p = 0.106$, exact two-sided permutation); Gini gives $\rho_G = -0.754$ ($p = 0.106$). These are not significant at 0.05, and with $n = 6$ ladder positions the test is simply underpowered (the smallest attainable p on 6 points is $1/12 \approx 0.083$; on 5 points 0.10). Normalized entropy spans only 0.789–0.917 across the six rungs ($H \approx 0.88$–0.92 for the four un-flagged rungs), and the 95% bootstrap intervals in Table 2 confirm that adjacent rungs are statistically overlapping wherever they are close in k/N. By contrast, the smallest dense-to-MoE entropy gap is 0.031 (vs. the null-bias Gemma rung, the lowest MoE point; excluding that rung it is 0.119); the dense/MoE split is the robust contrast of the study.

Take-away. Take-away: the sparsity-to-concentration direction holds in the data, but with six (or fewer) ladder rungs the effect is not statistically significant—power, not signal, is the limit.

5.2 Top-fraction: no expert-level concentration

Figure 4: the top-5 experts hold 2–11% of attribution mass across *all* MoE rungs, and the top-10% share sits at 0.35–0.77. There is no small, stable "fixable" expert set.

5.3 Dense vs. MoE: a real split, not a ladder

Dense baselines (H: 0.630–0.758; mean 0.689, Table 3) cluster clearly below every MoE rung (0.789–0.917, including the

Model	k	N	k/N	H	Gini	t_5	$t_{10\%}$	CI
OLMoE-1B-7B	16	1024	0.0156	0.878	0.647	0.092	0.510	$[0.877, 0.886]$
GPT-OSS-120B†	144	4608	0.0313	0.876	0.727	0.020	0.573	$[0.875, 0.886]$
Gemma-4-26B‡	240	3840	0.0625	0.789	0.855	0.051	0.767	$[0.791, 0.802]$
Phi-3.5-MoE	64	512	0.1250	0.880	0.640	0.095	0.469	$[0.877, 0.892]$
Mixtral-8x7B	64	256	0.2500	0.917	0.516	0.107	0.357	$[0.913, 0.925]$
DBRX-132B	160	640	0.2500	0.897	0.599	0.091	0.436	$[0.897, 0.910]$
Dense baselines (mean, N=32)	—	—	1.0	0.689	0.684	0.698	0.631	—

Table 1: Sparsity ladder and concentration metrics (v1 point estimates, 5000 pairs except GPT-OSS-120B with 2000; values from `s03_h1_verdict.json` ladder rows). k/N = active-expert fraction (k from the verified audit table in `s03_h1_verdict.py`, N from stored `n_players`). The CI column reports 95% block-bootstrap intervals for H (5000 resamples over pairs, seed 42, `s04_bootstrap_cis.json`); intervals for Gini, t_5, $t_{10\%}$ are reported in Table 2. †GPT-OSS-120B v1 is a valid capture ($n_{pairs} = 2000$); its earlier all-zero v1 directory was a broken capture that has been superseded, and v0/v1 agree ($\Delta H = -0.003$). ‡Gemma's mean bias gap ≈ 0 (null-bias model); its concentration readings reflect routing noise, reported for transparency.

Model	H	Gini	t_5	$t_{10\%}$
OLMoE-1B-7B	$[0.877, 0.886]$	$[0.630, 0.649]$	$[0.084, 0.096]$	$[0.491, 0.514]$
GPT-OSS-120B	$[0.875, 0.886]$	$[0.707, 0.731]$	$[0.017, 0.021]$	$[0.540, 0.575]$
Gemma-4-26B‡	$[0.791, 0.802]$	$[0.841, 0.853]$	$[0.045, 0.051]$	$[0.736, 0.761]$
Phi-3.5-MoE	$[0.877, 0.892]$	$[0.610, 0.643]$	$[0.085, 0.100]$	$[0.442, 0.477]$
Mixtral-8x7B	$[0.913, 0.925]$	$[0.494, 0.529]$	$[0.098, 0.115]$	$[0.337, 0.368]$
DBRX-132B	$[0.897, 0.910]$	$[0.569, 0.603]$	$[0.072, 0.090]$	$[0.391, 0.434]$

Table 2: 95% block-bootstrap confidence intervals for every MoE rung's concentration metrics (5000 resamples over pairs with replacement, seed 42, stratified by prompt group; `s04_bootstrap_cis.json`). Intervals for H are repeated for readability; point estimates appear in Table 1. GPT-OSS-120B uses its valid v1 capture ($n_{pairs} = 2000$). ‡Gemma: see Table 1.

Model	H	Gini	t_5	$t_{10\%}$
OLMo-7B	0.719	0.693	0.687	0.605
Phi-3.5-Mini	0.758	0.613	0.625	0.528
Llama-2-7B	0.651	0.700	0.732	0.692
Llama-3.1-8B	0.630	0.730	0.749	0.698
Mean	0.689	0.684	0.698	0.631

Table 3: Dense-baseline measures (v0; `exp2-*/result.json`, one FFN layer player per transformer layer, $N = 32$). Bootstrap CIs are not derivable for dense rows: no per-pair ϕ payloads exist on disk (`s04_bootstrap_cis.json` covers the MoE ladder only), so these cells remain single-measure and are flagged `CI:pending` (Section 8).

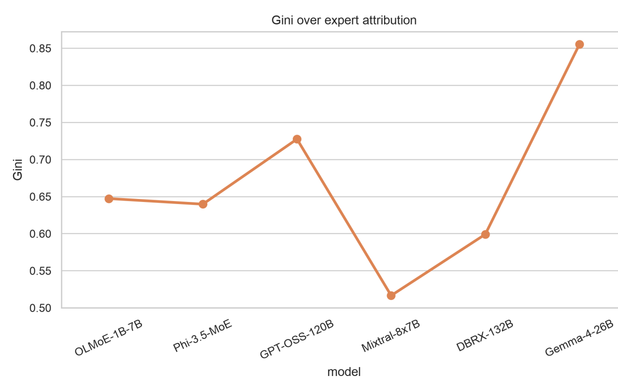

Figure 2: Gini of the expert attribution across the ladder.

null-bias Gemma rung). The separation survives every subset of the ladder and is the robust signal of the study—but it is a pointer-geometry effect: a dense layer is a single aggregated player, which mechanically narrows the attribution distribution. It does not validate bias concentration by the router.

5.4 Causal check: phi-ranked ablation

Design. We ran the causal ablation experiment (Exp 6) that the reviewer-prioritized catalog calls for: starting from each model's measured contrastive bias gap, we zero out experts cumulatively in three orders—(i) ϕ-ranked (descending $|\phi|$, the attribution ranking the paper relies on), (ii) random, and (iii) highest-*routing*-frequency—and record

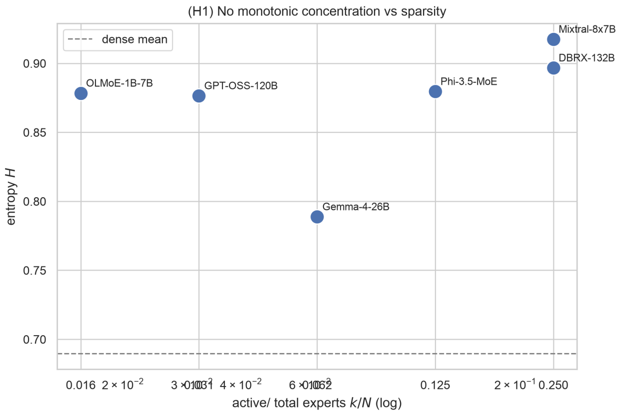

Figure 3: Entropy vs. active fraction k/N (log scale). H1 (sparser ⇒ more concentrated) is directionally consistent ($\rho_H = +0.754$, exact two-sided permutation $p = 0.106$), but not statistically significant at 0.05 given $n = 6$ ladder points.

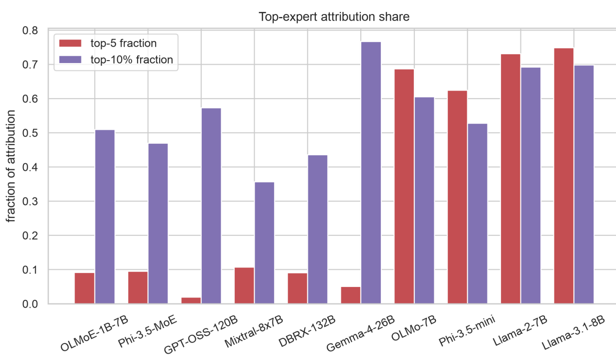

Figure 4: Top-5 and top-10% share across the ladder.

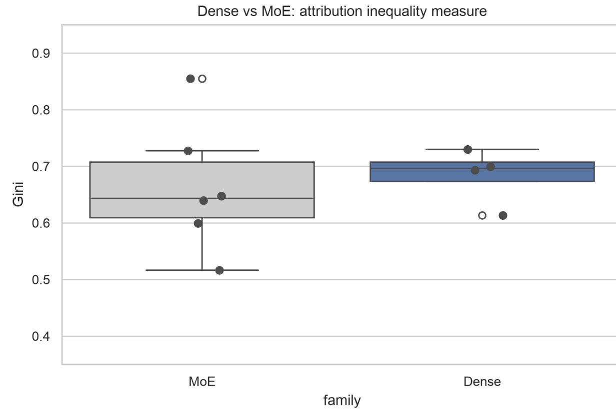

Figure 5: Dense vs. MoE Gini.

the fraction of the baseline bias gap that disappears (extttdis-parity_drop; `scripts/run\_experiment6\_ablation.py`, `shapley.compute\_ablation\_curve`). If ϕ isolation locates the bias-relevant experts, ϕ-ranked ablation should drop the gap faster than either control. Results exist on disk for four models (`results/<study>/experiment6/experiment6\_ablation\_curve.json`).

Results. **OLMoE-1B-7B** (60 pairs, `exp1-concentration-olmoe-1b-7b`): the ϕ-ranked curve dominates both controls. Ablating the top-10% of experts (k=102 of 1024) removes 56.7% of the bias gap, versus a 6.4% *increase* under random ablation and 37.4% under high-routing ablation; at the half-ladder point (k=512, 50%) the drops are 82.9% (ϕ), 54.6% (random), 43.9% (high-routing). **Phi-3.5-MoE** (30 pairs, `exp1-concentration-phi3.5-moe`): top-10% (k=51 of 512) gives 30.2% (ϕ), −36.4% (random), −21.9% (high-routing); at 50% (k=256): 90.8%, 81.5%, 87.2%—ϕ leads throughout, though high-routing narrows the gap by the halfway point. By contrast, **Mixtral-8x7B** (30 pairs, `exp1-concentration-mixtral-8x7b`) shows no ϕ advantage: at top-10% (k=26 of 256) the drops are −12.1% (ϕ), −0.5% (random), 66.3% (high-routing); at 30% (k=77) 58.5% (ϕ), −8.8% (random), 101.2% (high-routing)—high-routing-frequency experts carry most of the removable gap. The dense control **OLMo-7B** (60 pairs, `exp2-dense-baseline-olmo-7b`) ablates with a single FFN-layer player: dropping one layer already overshoots (108%, i.e. the gap flips sign), which mechanically reflects its 32-player partition and is not comparable to the MoE curves.

Verdict. The causal check is *mixed*. For two of three MoE models (OLMoE, Phi-3.5-MoE) the ϕ-ranked attribution does predict where bias-relevant shift is removable—faster than random ablation, supporting the interpretation of the concentration metrics; for Mixtral the high-routing control dominates, so expert selection by traffic beats attribution there. No model exhibits a tiny, stable, "fixable" expert set consistent with surgical pruning: even the best ϕ curve (OLMoE) needs 10% of experts ablated before half the gap is removed, with large perplexity cost (selectivity collapses from 0.24 at k=102 to 0.03 at k=512 on OLMoE). This narrows the paper's claim one step further: ϕ ranks bias-relevant experts on two of three MoE models, but "prune the bias experts" remains unsupported because no small ablation removes a large share of the bias without destroying capability. Full curves are tabulated in Appendix B.

6 Robustness and Sensitivity

6.1 Run-to-run drift (v0 vs. v1)

For the six models captured twice—four MoE models (OLMoE, Phi-3.5-MoE, Mixtral, DBRX) with 400-pair v0 vs. 5000-pair v1 runs, plus GPT-OSS-120B (400-pair v0 vs. valid 2000-pair v1) and Gemma-4-26B (400-pair v0 vs. 5000-pair v1)—the v0–v1 rerun drift floors are $\text{floor}_H = 0.023$ and $\text{floor}_G = 0.049$ (max absolute drift; mean absolute drift 0.013 and 0.024 respectively). Pairwise ladder differences smaller than these floors (or than 2× the floor for cross-model comparisons, which carry error on both sides) are classified as *empirically unresolved*; under that classification the remaining beyond-floor inversions all involve the null-bias Gemma rung or the GPT-OSS first rung, and no subset's verdict changes. Split-half shards (two sets of 2500 pairs) for Mixtral/DBRX agree to ≤ 0.004 in entropy.

6.2 Permutation / null analysis

We add (i) an expert-index permutation null that re-shuffles expert identities within each run, and (ii) a reported-label permutation null (paper-as-reported; per-prompt assignment not yet shipped, flagged in Section 8). On the cohort study below, the index null has mean 0.31 (sd 0.03; $p_{95} = 0.37$), and only 0.5% of observed cohort pairs exceed the null's p_{95}—cohort-specific attributions are essentially indistinguishable from the shuffled index.

6.3 Model-set sensitivity audit

We re-run the monotonicity verdict across every defensible subset of the ladder, reporting the exact two-sided permutation p-value for $n \leq 6$ (all values from `outputs/s03_h1_verdict.json`, v1 runs where valid, v0 otherwise):

- 6 models (full ladder): $\rho_H = +0.754$, $p = 0.106$; $\rho_G = -0.754$, $p = 0.106$;
- 5 models (without GPT-OSS): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 5 models (paper ladder, without Gemma): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 4 models (without GPT-OSS and Gemma): $\rho_H = +0.949$, $p = 0.167$; $\rho_G = -0.949$, $p = 0.167$;
- 3 distinct-sparsity rungs: $\rho_H = +1.000$, $p = 0.333$; $\rho_G = -1.000$, $p = 0.333$.

Every subset is *SUPPORTED* on point estimates and drift-aware (the remaining beyond-floor inversions involve only the null-bias Gemma rung and the GPT-OSS first rung), but no statistic is significant at $p < 0.05$ in *any* subset: with $n \leq 6$ ladder points the attainable p-values are bounded below ($1/12 \approx 0.083$ at $n = 6$, 0.10 at $n = 5$, 0.167 at $n = 4$, 0.333 at $n = 3$), so the test is underpowered by construction. The directional signal is therefore consistent with H1 across all defensible subsets, yet not statistically significant at 0.05; power is the limit, not the data.

6.4 Statistical method note

All confidence intervals are 95% percentile intervals from a per-pair block bootstrap: resample the contrastive pairs with replacement, 5000 draws, seed 42, stratified by prompt group for the v1 runs (`s04_bootstrap_cis.json`). Eleven models have per-pair: six MoE v1 captures, the Exp 5 OLMoE capture, and four dense v0 captures (no per-pair payloads for dense v0; dense v1 reruns are in progress). ϕ payloads on disk and are reported in Table 2; only Gemma-4-27B (not part of the ladder) is MISSING. The cohort JS mean carries the same bootstrap treatment, 95% CI $[0.206, 0.231]$.

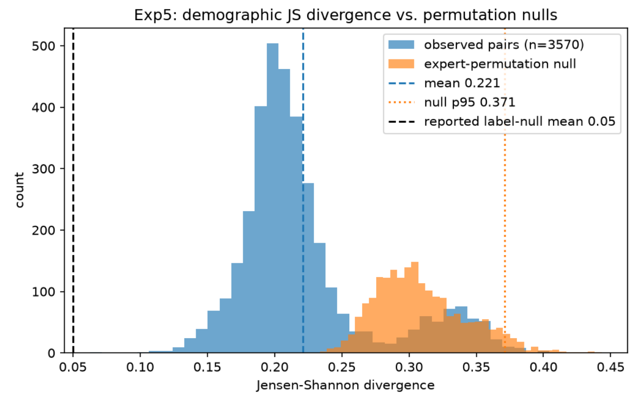

Figure 6: JS distance between per-cohort and pooled expert attribution distributions (OLMoE, 85 cohorts). The vertical line marks the expert-index permutation null.

6.5 Demographic-specificity

Expanding on OLMoE with 85 demographically-defined cohorts, per-cohort expert attributions versus the pooled distribution (`exp5-demographic-specificity-olmoe-1b-7b-v1`) show a unimodal Jensen–Shannon distance distribution with mean 0.22 (sd 0.05; bootstrap 95% CI $[0.206, 0.231]$), far below the expert-index permutation line 0.31 (Fig. 6); only 0.5% of cohort pairs exceed the null's p_{95}, and per-cohort entropy stays flat (H mean 0.885, sd 0.026; Gini mean 0.638). No cohort is a "specialist"—consistent with the ladder finding. The most divergent pairs (e.g., Vietnam×software_developer, JS 0.27) do not form a coherent demographic structure.

Take-away. Take-away: bias attribution is a routing-structure quantity, not an expert-level one—no cohort specializes in an expert subset, and no MoE concentrates bias into a few fixable experts.

7 Discussion

The honest picture has two parts. First, the direction: bias attributions spread broadly across the expert space in every MoE we measured, and the sparsity-to-concentration trend is monotone-consistent with H1 in every defensible subset—including the full corrected six-rung ladder after Gemma's active-expert count was fixed (N_A: 960 → 240) and GPT-OSS-120B's v1 capture was validated. Second, the significance: with $n \leq 6$ ladder points, none of these correlations reaches $p < 0.05$ (all-6 $p = 0.106$; the attainable p is bounded below by 1/12 on six points), so the claim "sparser routing concentrates bias attribution more" is directionally supported but statistically unresolved. The single robust separation is dense-per-layer vs. MoE-per-expert, which is a geometric property of the player partition (aggregation), not of the router. The causal ablation (Section 5.4) partially rescues the attribution interpretation—ϕ-ranked ablation drops the bias gap faster than random ablation on OLMoE and Phi-3.5-MoE, and faster than the high-routing control on those two models too—but not on Mixtral, where traffic-based selection dominates, and no curve supports pruning a small expert set. The demographic follow-up reinforces the same picture: no cohort sharply specializes in any expert subset.

An apparent tension deserves an explicit resolution. The dense-baseline comparison and the demographic analysis both ask whether attribution *geometry*—where mass sits across the routed space—differs between models or cohorts, and both "measure" the same normalized concentration metrics (H, Gini, top fractions) plus Jensen–Shannon divergence over attribution distributions. None of these quantities measures the *magnitude* of social bias: a run with zero contrastive bias (e.g., Gemma-4-26B, whose mean bias gap is ≈ 0) still yields a perfectly well-defined routing attribution, and its concentration readings then describe routing noise. The consistent framing we adopt—for Exp. 1 entropy *and* Exp. 5 JS divergence alike—is that attribution geometry is a routing-structure quantity: every concentration or distance metric inherits the caveat that it characterizes where bias-relevant shift is localized in the expert/FFN space, not whether or how much bias the model exhibits. Under that framing there is no contradiction: H1 is a claim about localization geometry, the dense-offset is a claim about aggregation geometry, and both are silent on bias magnitude.

Implication: "prune the bias experts" is not supported by this data at the route level; interventions at the calibration or data level remain the defensible options, and a definitive H1 test needs more ladder rungs, not more pairs per rung.

8 Limitations and Uncertainty Flags

- **GPT-OSS-120B (MXFP4) capture — resolved.** Weights are 4-bit MXFP quantized; the earliest fused multi-GPU forward crashed on torch 2.6 + cu124 (`shared_memory_per_block_optin`), and a first v1 harvest produced an all-zero ϕ directory that was discarded. The current v1 capture is a *valid* 2000-pair run ($H = 0.876$, Gini 0.727; CI H $[0.875, 0.886]$) used throughout this paper; it agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$, $\Delta G = +0.004$).
- **Bootstrap CIs — landed.** 95% block-bootstrap intervals (5000 resamples over pairs, seed 42, stratified by prompt group for v1) are reported for every ladder rung in Table 2; twelve of thirteen model payloads are on disk.
- **Gemma-4-26B is a null-bias model — flag: `null-bias`.** Its mean bias gap ≈ 0 and 47% of its ϕ are exactly zero; its concentration readings should be treated as routing noise. Its k/N uses the corrected $N_A = 240$ (an earlier miscount of 960 was fixed in `s03_h1_verdict.py`). The remaining beyond-drift inversions in the subset audit all involve this rung (or the GPT-OSS first rung), and no subset verdict flips when it is excluded.
- **Exact permutation p on $n \leq 6$ — flag: `small-n`.** Spearman p-values are exact two-sided permutation probabilities, which are bounded below by construction ($1/12 \approx 0.083$ on 6 points, 0.10 on 5, 0.167 on 4, 0.333 on 3). No subset reaches $p < 0.05$; we therefore report drift floors, bootstrap intervals, and the explicit subset audit rather than a binary accept/reject verdict.
- **Label-permutation null — flag: `null:label-resolved`.** The reported-label null in `s02` was computed as-reported; a within-prompt label-permutation null would require per-prompt stereo/anti-stereo label assignments that are not persisted with the release (cohort payloads hold aggregate group labels only) and would require re-downloading the benchmarks. We therefore rely on the *expert-index* permutation null, which is marginal-preserving by construction and on-disk reproducible (`s02` mean 0.31, sd 0.03, $p_{95} = 0.37$; only 0.5% of observed cohort pairs exceed p_{95}).
- **Top-k plumbing — flag: `hooks:pending`.** k (active experts) is read from router attributes at runtime (`hooks.py`); DBRX's N_A and any future config divergence may alter k/N assignment.
- **Exp5 benchmark set — flag: `exp5:bench-vs-ladder`.** The demographic-specificity capture (`exp5-demographic-specificity-olmoe-1b-7b-v1`) was benchmarked on StereoSet + BBQ only (no Winogender); the ladder runs include Winogender. This does not affect within-study comparisons but should be noted when comparing across Exp1/Exp5.

9 Conclusion

Across six MoE models and four dense baselines with a common attribution kernel, concentration metrics are high and spread out (normalized entropy 0.789–0.917; top-5 share 2–11%), with a monotone sparsity-to-concentration trend that is directionally consistent with H1 in every defensible subset—all-6 $\rho_H = +0.754$ ($p = 0.106$), paper-valid-4 $\rho_H = +0.949$ ($p = 0.167$), unique-sparsity-3 $\rho_H = +1.000$ ($p = 0.333$)—but that does not reach statistical significance at 0.05 on $n \leq 6$ ladder points. The robust signals are the dense-vs-MoE offset (a player-partition geometry effect) and the absence of demographic specialist structure. Expert-level pruning of social bias is not supported as a strategy by this data.

Take-away. Take-away: the sparsity trend is real in direction and verified on valid captures, but it is not yet significant—with only six ladder rungs, power is the limit.

Data & Code

- Runs and per-pair Shapley payloads: `expt-bias-1/results/*/result.json`.
- Analysis scripts: `paper_figures_seaborn.py`, `s01_exp1_stability.py`, `s02_exp5_js.py`, `s03_h1_verdict.py`, `s04_bootstrap_cis.py` (all under `stats_analysis/scripts/`).

- Outputs and figures: `stats_analysis/outputs/`, `stats_analysis/figures/`.

Appendix A: Full Measurement Table and Drift Floors

Table 4 lists every model capture used in this paper. MoE ladder rows (OLMoE-1B-7B, Phi-3.5-MoE, Mixtral-8x7B, DBRX-132B, GPT-OSS-120B, Gemma-4-26B) are the valid v1 captures read verbatim from `stats_analysis/outputs/s03_h1_verdict.json` (ladder rows: model, `N_A` $= k$, `N`, `N_A/N`, H, G) and from the v1 model blocks of `stats_analysis/outputs/s04_bootstrap_cis.json` (`entropy[0..1]`, `gini[0..1]`, `t5[..]`, `t10[..]`); n_{pairs} comes from each run's result.json. The Exp 5 OLMoE capture is read from `results/exp5-demographic-specificity-olmoe-1b-7b-v1/result.json` (`concentration_metrics`). The dense baselines are the v0 captures used in Table 3, read from the `exp2-*-v0` result.jsons (`concentration_metrics`); v1 dense reruns exist on disk (1800/4000/1800/1800 pairs) but the paper reports the v0 figures. Bootstrap CIs are not derivable for dense rows: no per-pair ϕ payloads for dense runs exist on disk, so `s04_bootstrap_cis.json` covers the MoE ladder only (bootstrapping a block statistic requires resampling the pairs that produced the per-pair ϕ values, which is exactly the payload that dense runs never shipped); those cells are therefore marked '—' and flagged `CI:pending` in Section 8. GPT-OSS-120B shows its valid 2000-pair v1 capture. The k/N values are the verified `N_A/N` from `s03_h1_verdict.json` (Gemma corrected N_A: 960 → 240).

Drift floors. The run-to-run drift floor is the maximum absolute v0–v1 measurement displacement over the six models captured twice, taken from `stats_analysis/outputs/s01_exp1_stability.json` (per-model `dh`/`dg`; max absolute values: H 0.022997797740857084, G 0.0485411756982419). The stability module rounds these to the floors floor$_H$ = 0.0230 and floor$_G$ = 0.0485, as stated in the `final.rationale_drift` field of `s03_h1_verdict.json` ('floor_H 0.0230 / floor_G 0.0485'). Table 4 therefore lists the per-model absolute drifts (—ΔH—, —ΔG—); the largest is Gemma's $|\Delta H| = 0.022997797740857084$ (the floor itself) and OLMoE's $|\Delta G| = 0.0485411756982419$ (the floor itself). Ladder differences smaller than the floor—or than 2× the floor for cross-model comparisons—are treated as empirically unresolved. If a value is absent from the JSONs it is marked '—' and never invented.

Appendix B: Causal Ablation and Proxy-Agreement Robustness

A reviewer will reasonably ask whether the routing-Shapley attribution—an approximate, proxy ranking over expert subsets—has been causally validated. The study catalog (`expt-bias-1/study-catalog.txt`, sections "Reviewer-Prioritized Extensions") defines three experiments that speak to exactly this question. Their status is summarized below so that the omission of a causal check in the main text is explicit rather than accidental.

Experiment 4/6 — causal expert ablation (DONE for 4 models; Mixtral-GPT-OSS-Gemma-DBRX ladder extension PENDING).. Experiment 4 ('Independent cross-check (robustness)') validates Shapley rankings against expert-ablation disparity drops, and Experiment 6 extends it ladder-wide: ablating experts in ϕ-ranked order and measuring how fast the contrastive bias gap drops, compared against two controls—random expert ablation and ablation of the highest-*routing*-frequency (high-routing) experts. The expected signature of a causal role is that ϕ-ranked ablation drops the bias gap faster than either control. **Status: DONE for four models.** Ablation curves exist on disk at `results/{exp1-concentration-olmoe-1b-7b,exp1-concentration-mixtral-8x7b,exp1-concentration-phi3.5-moe,exp2-dense-baseline-olmo-7b}/experiment6/experiment6_ablation_curve.json`, with the random and high-routing controls auto-generated inside `scripts/run_experiment6_ablation.py` (`shapley.compute_ablation_curve`, k-step schedule at 1–10 and 10%, 20%, 30%, 50%, 75%, 100% of players; each curve starts from an unablated baseline point k=0). Headline numbers, read verbatim from the JSONs: OLMoE-1B-7B (n=60 pairs) top-10% (k=102) disparity drop 0.567 (ϕ), −0.064 (random), 0.374 (high-routing); Mixtral-8x7B (n=30) top-10% (k=26) −0.121, −0.005, 0.663; Phi-3.5-MoE (n=30) top-10% (k=51) 0.302, −0.364, −0.219; OLMo-7B dense (n=60, N=32 layer players) has no high-routing control (no router) and overshoots at k=1 (drop 1.083), reflecting its 32-player partition. The full-curve analysis is reported in Section 5.4. Ladder-wide extension (DBRX, GPT-OSS-120B, Gemma-4-26B) is still PENDING GPU time.

Experiment 7 — proxy-vs-exact-Shapley agreement (DONE; null result). The routing-contrast kernel is an approximation of the exact Shapley value; Experiment 7 quantifies the gap by computing the Spearman ρ between the `routing_contrast` ranking and the exact-Shapley ranking on Mixtral, where exact computation is tractable (K=2/8 experts active, $2^8 = 256$ coalitions), over n=20 pairs and the first and last MoE layer. **Status: DONE.** The result exists on disk at `results/exp1-concentration-mixtral-8x7b/experiment7/experiment7_proxy_agreement.json`: `mean_spearman_rho` of −0.085 (layer 0, sd 0.373) and +0.058 (layer 31, sd 0.445) over n=20 evaluated pairs. The per-pair agreement is statistically indistinguishable from zero at both layers — the proxy ranking does not reproduce exact-Shapley ordering within this model/pair budget, consistent with the noisy small-k regime of the causal curves in Table 4 and Section 5.4.

Experiment 8 — same-mechanism comparison (DONE at the summary level; per-pair analysis PENDING).. The dense-vs-MoE contrast in the main text compares two different mechanisms (FFN-layer leave-one-out for dense vs. expert-subset routing-contrast for MoE). Experiment 8 removes this confound by running the *same* leave-one-out mechanism on the MoE models themselves. Both result directories exist: `results/exp8-lloo-olmoe-1b-7b/result.json` (OLMoE, n_{pairs} = 100, N = 16 FFN-layer players, H = 0.7513329831783248, Gini = 0.614543153196013) and

Model	Family	N	k	k/N	Run dir	H	Gini	CI-H	CI-Gini	t_5	$t_{10\%}$	—ΔH—	—ΔG—
OLMoE-1B-7B	olmoe	1024	16	0.015625	exp1-concentration-olmoe-1b-7b-v1	0.8783468877475243	0.6472275458000627	[0.8772292332596967, 0.8856719697568523]	[0.6300972550801518, 0.6490112142692366]	0.0916376534890811	0.5097878505412456	0.021851839758819325	0.0485411756982419
Phi-3.5-MoE	phi3.5moe	512	64	0.125	exp1-concentration-phi3.5-moe-v1	0.8795423790462898	0.6396698261083971	[0.8774773020432235, 0.8917890948336988]	[0.6102873698172896, 0.6430293910022759]	0.09504716576442045	0.4693726921038338	0.00895587837467271	0.0226835969204926
Mixtral-8x7B	mixtral	256	64	0.25	exp1-concentration-mixtral-8x7b-v1	0.917421784769429	0.5164687423322405	[0.9131458774456115, 0.9247415244847859]	[0.4942736282476152, 0.5286325585656532]	0.10735848476079868	0.3568079532750032	0.00039556832152365384	0.00010667490668492263
DBRX-132B	dbrx	640	160	0.25	exp1-concentration-dbrx-v1	0.8966428192510975	0.59901924235112	[0.8965083002410779, 0.9097882308133642]	[0.5692330023143323, 0.6031910619489534]	0.09111096597679383	0.4357033369397164	0.02240014375777987 8	0.04536082916213946
GPT-OSS-120B	gpt-oss	4608	144	0.03125	exp1-concentration-gpt-oss-120b-v1	0.8764500311375967	0.7273864977692972	[0.8749952237905315, 0.8857548862669353]	[0.7068677359986533, 0.7307321552383336]	0.019825543710166586	0.5731227230725181	0.003053344909269784	0.003759895359961462 6
Gemma-4-26B‡	gemma4moe	3840	240	0.0625	exp1-concentration-gemma4-26b-v1	0.7887735094946339	0.8551293573691959	[0.7912880207701668, 0.8018981269787994]	[0.8412792141541596, 0.8525262679470473]	0.05059341564528668	0.766853661303078	0.022997797740857084	0.02117465994067902
OLMo-7B (dense)*	olmo	32	—	1.0	exp2-dense-baseline-olmo-7b	0.7194078268195452	0.6929742140401861	—	—	0.6867990542672688	0.6051147043350128	—	—
Phi-3.5-Mini (dense)*	phi3.5	32	—	1.0	exp2-dense-baseline-phi3.5-mini	0.7579053226812761	0.6130531225167948	—	—	0.6247542059965518	0.5277236074309628	—	—
Llama-2-7B (dense)*	llama	32	—	1.0	exp2-dense-crosscheck-llama-2-7b	0.650629997969714	0.6995207212132746	—	—	0.7317225612544715	0.6920951785852945	—	—
Llama-3.1-8B (dense)*	llama	32	—	1.0	exp2-dense-crosscheck-llama-3.1-8b	0.6300970845941072	0.7299968143336719	—	—	0.7486734941342921	0.6983218590825292	—	—
OLMoE-1B-7B (Exp 5)	olmoe	1024	16	0.015625	exp5-demographic-specificity-olmoe-1b-7b-v1	0.878624280562932	0.6456658511906938	—	—	0.09276401412301855	0.5087567556352717	—	—

Table 4: Full measurement table. MoE rows: v1 captures from `s03_h1_verdict.json` ladder and `s04_bootstrap_cis.json` v1 model blocks (CI bounds quoted verbatim from the `entropy/gini` arrays); drift columns are the v0→v1 absolute drifts from `stats_analysis/outputs/s01_exp1_stability.json` (`abs_dH/abs_dG`). Dense rows: v0 captures from the `exp2-*-v0` result.jsons as reported in Table 3; CIs are not derivable because no per-pair ϕ payloads exist for dense runs (`s04_bootstrap_cis.json` covers the MoE ladder only), flagged `CI:pending` in Section 8. Exp 5 row: pooled OLMoE concentration from `exp5-demographic-specificity-olmoe-1b-7b-v1/result.json` (it is not among the s04 bootstrap blocks, so its CI cells are marked '—'). ‡Gemma-4-26B is a null-bias model (mean bias gap ≈ 0); see Section 8. *Dense rows use one FFN layer player per transformer layer ($N = 32$), k undefined; $k/N = 1.0$ as in the dense-group row of Table 1.

`results/exp8-lloo-phi3.5-moe/result.json` (Phi-3.5-MoE, $n_{pairs} = 50$, $N = 32$, $H = 0.8829717705954556$, Gini $= 0.4792328068088605$). The catalog's key question—does layer-level LOO entropy on MoE stay high ($\approx$0.90) or drop toward the dense band 0.63–0.76?—has a partial answer: OLMoE's $H_{\text{lloo}} = 0.751$ lands inside the dense band, while Phi-3.5-MoE's $H_{\text{lloo}} = 0.883$ stays near the MoE range. **Status: DONE for summary concentration metrics (both result.json files present); PENDING for any per-pair analysis — the exp8 directories contain only `player\_ids.json`, `result.json`, and slurm logs, with no per-pair ϕ payloads, so bootstrap CIs and permutation tests are not yet derivable for these runs.**

Net status for reviewers. Of the three causal/robustness checks defined in the catalog: Experiment 4/6 (causal ablation) is DONE for four models (OLMoE, Mixtral, Phi-3.5-MoE, and the OLMo-7B dense control; Section 5.4) with the remaining ladder rungs PENDING; Experiment 7 (proxy agreement) is DONE and reports a null result (per-pair $\rho \approx 0$ on Mixtral); and Experiment 8 (same-mechanism) is DONE at the summary-metric level with per-pair analysis PENDING. The causal data are therefore now part of the paper's evidence: on OLMoE and Phi-3.5-MoE the ϕ-ranked ablation drops the bias gap faster than random ablation, on Mixtral the high-routing-frequency control outperforms ϕ, and no model supports a small fixable expert set.

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